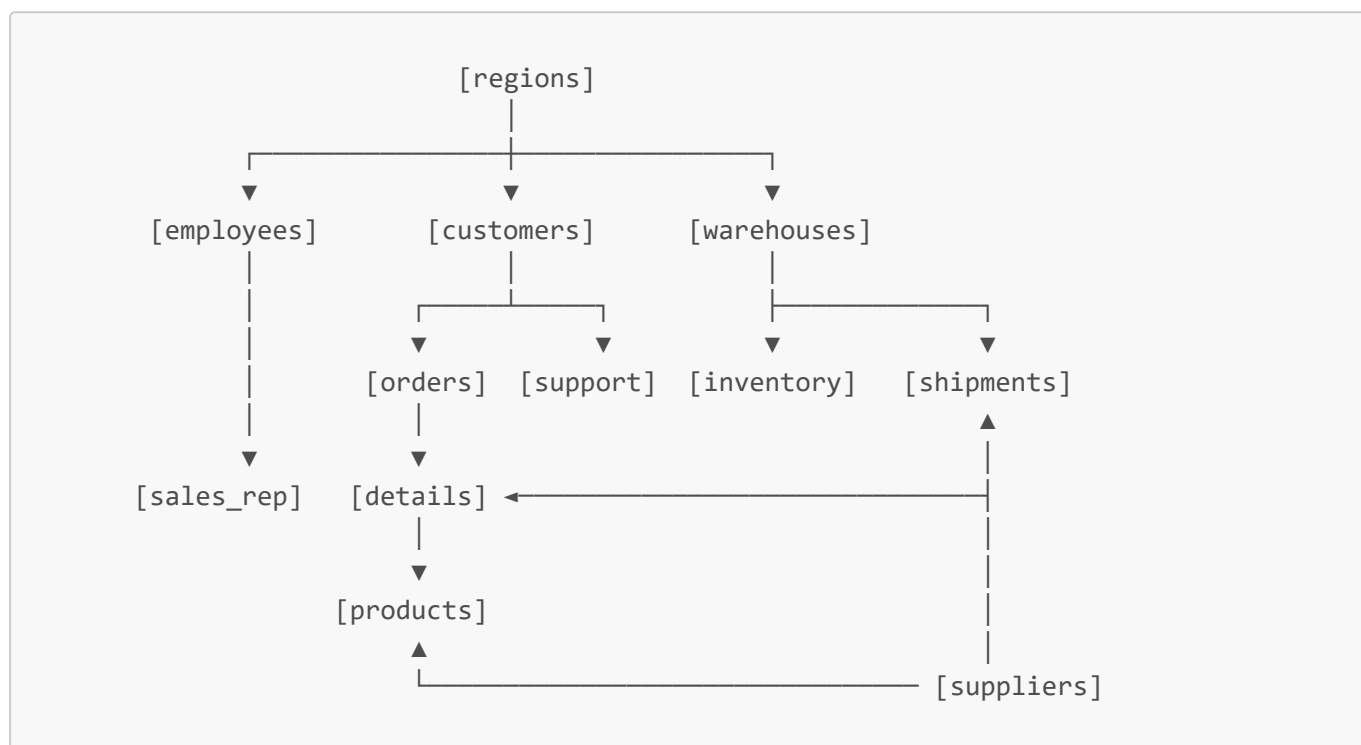


DecisionIQ Master Handbook

CHAPTER 8: Data Engineering & Relational Warehouse Schema Design

8.1 Data Schema Layout

DecisionIQ uses a normalized Snowflake schema to structure lookup and transactional data tables:



8.2 Schema Table Profiles

- **Dimensional/Lookup Tables:**
 - **regions**: Sales territories.
 - **employees**: Personnel reports_to chains.
 - **customers**: B2B client details.
 - **products**: Catalog software/hardware configurations.
 - **warehouses**: Physical inventory holding hubs.
 - **suppliers**: Parts providers.
 - **marketing_campaigns**: Ad spend budgets and conversion metrics.
- **Transactional/Fact Tables:**
 - **orders**: Transaction headers.
 - **order_details**: Granular invoice lines.
 - **payments**: Financial settlements.
 - **shipments**: Logistics delivery transactions.
 - **customer_support**: Support ticket lifecycles and CSAT scores.
 - **daily_business_metrics**: Web traffic and NPS aggregates.

8.3 Constraint Design

We enforce strict checks and constraints at the database storage layer to preserve data quality:

- `UNIQUE(warehouse_id, product_id)` in the `inventory` table prevents duplicate stock lines.
 - `CHECK (unit_price >= 0)` in `products` protects pricing ranges from garbage inputs.
 - `CHECK (reliability_score >= 0.0 AND reliability_score <= 1.0)` in `suppliers` keeps scoring metrics within bound.
-

8.4 Indexing Strategy

To avoid slow table scans on dashboard loads, we place B-Tree indexes on joined columns:

```
CREATE INDEX idx_orders_customer_id ON orders(customer_id);
CREATE INDEX idx_orders_order_date ON orders(order_date);
CREATE INDEX idx_order_details_order_id ON order_details(order_id);
```

Indexes speed up joins from $O(N)$ table scans to $O(\log N)$ tree traversals.